

VSP5600 Performance Analysis Report

Storage Performance & Capacity Analysis

Analysis Date: 2024-12-30

Performance Data Date: 2024-12-30

System: Hitachi VSP5600 Storage Array

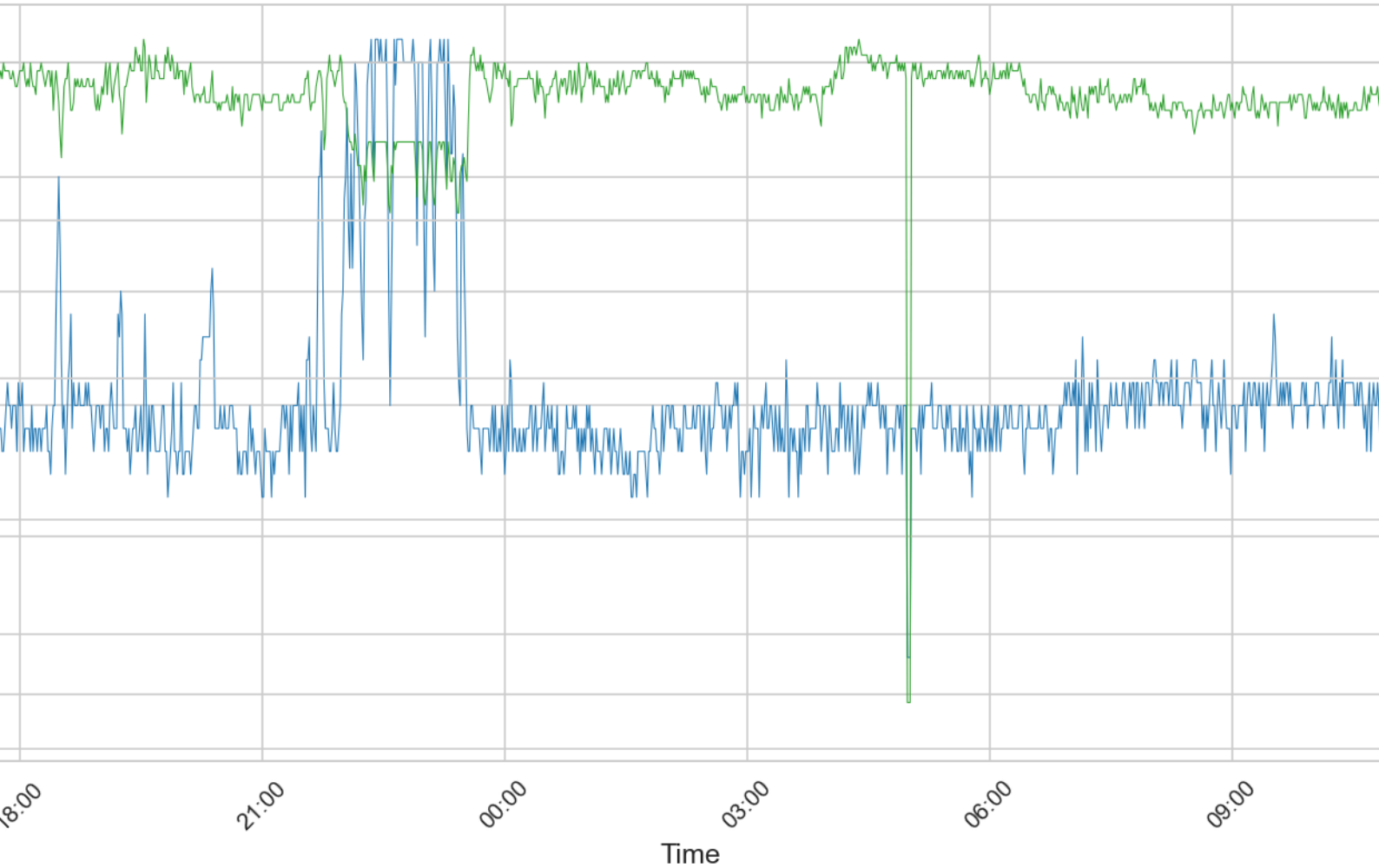
Report Generated By: Performance Analysis System

This report provides a comprehensive analysis of storage performance metrics collected from the Hitachi VSP5600 system. It includes detailed charts and analysis of cache performance, processor utilization, port response times, and LDEV (Logical Device) metrics.

6.1 Cache Metrics

Write Pending Rate vs Cache Usage Rate

Date: 2025/12/18



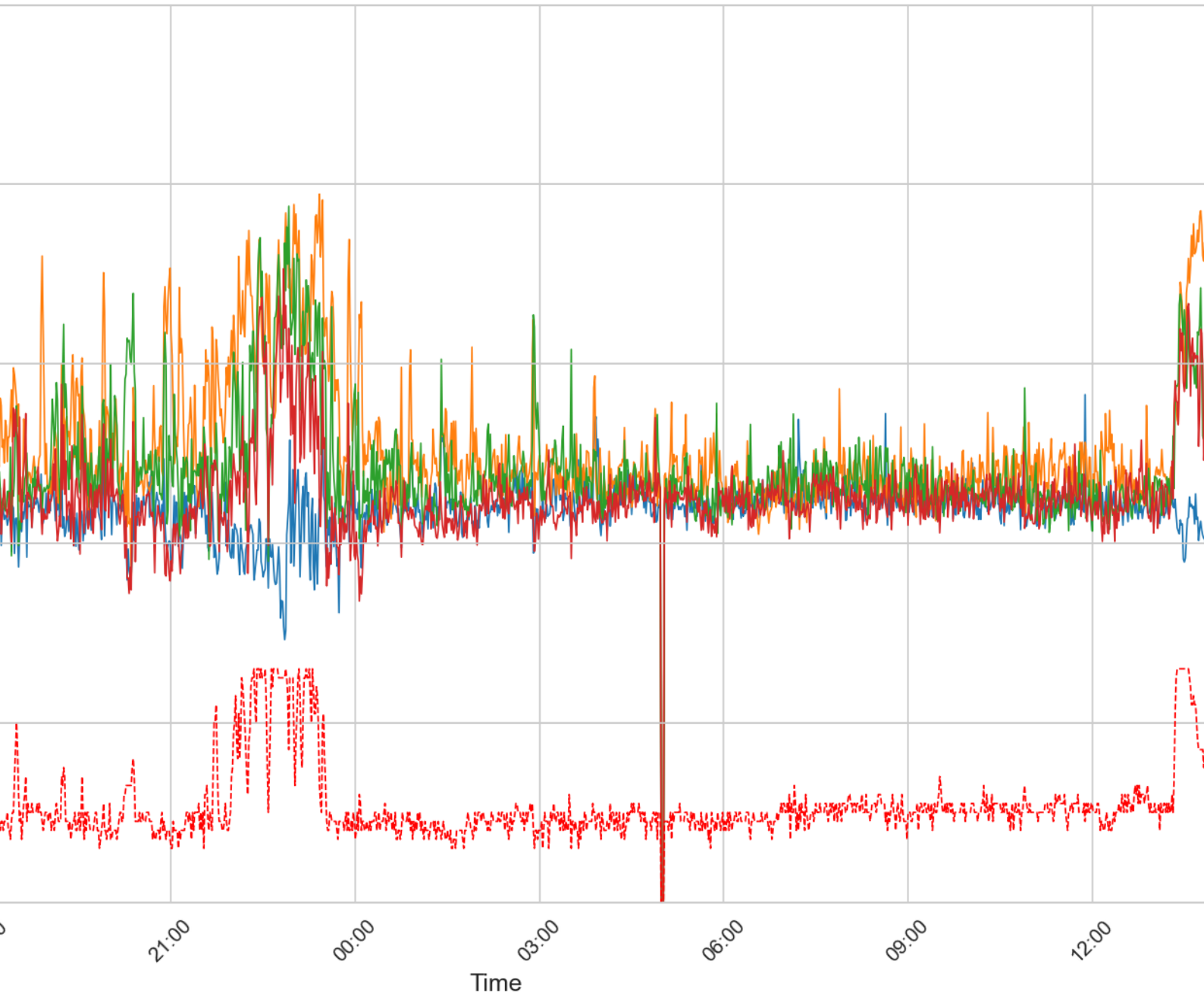
Analysis:

Cache write pending seems to peak to 25% between 21:00 and 00:00. Check the MPU usage at this time and identify the LDEVs that have high transfer rates and IOPS.

6.2 MPU Usage vs Write Pending Rate

MPU Usage vs Write Pending Rate

Date: 2025/12/18



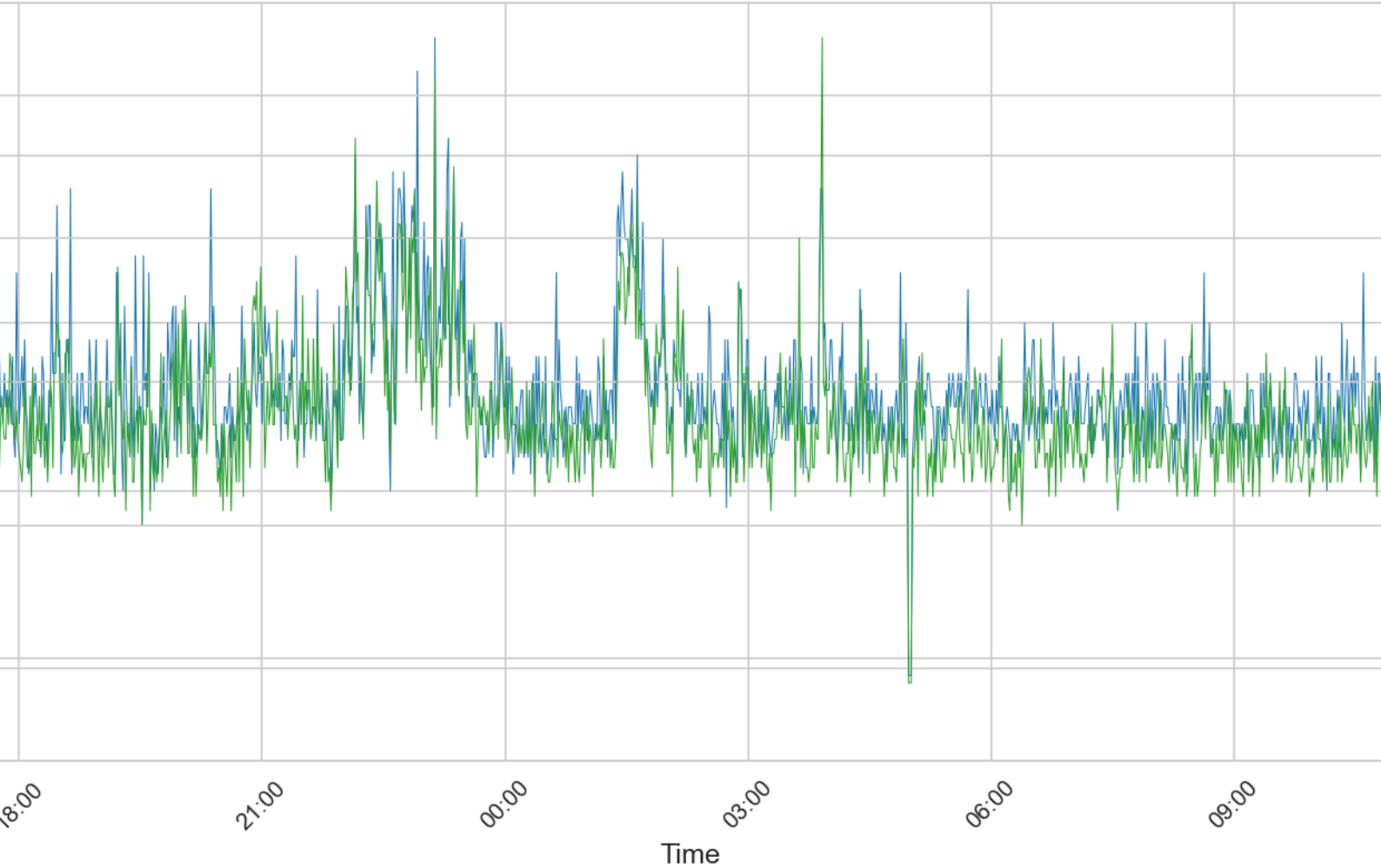
Analysis:

MPU usage is high when write pending is high between 21:00 and 00:00.

6.3 HIE Metrics: HIE ISW vs MPU HIE

HIE ISW vs MPU HIE

Date: 2025/12/18



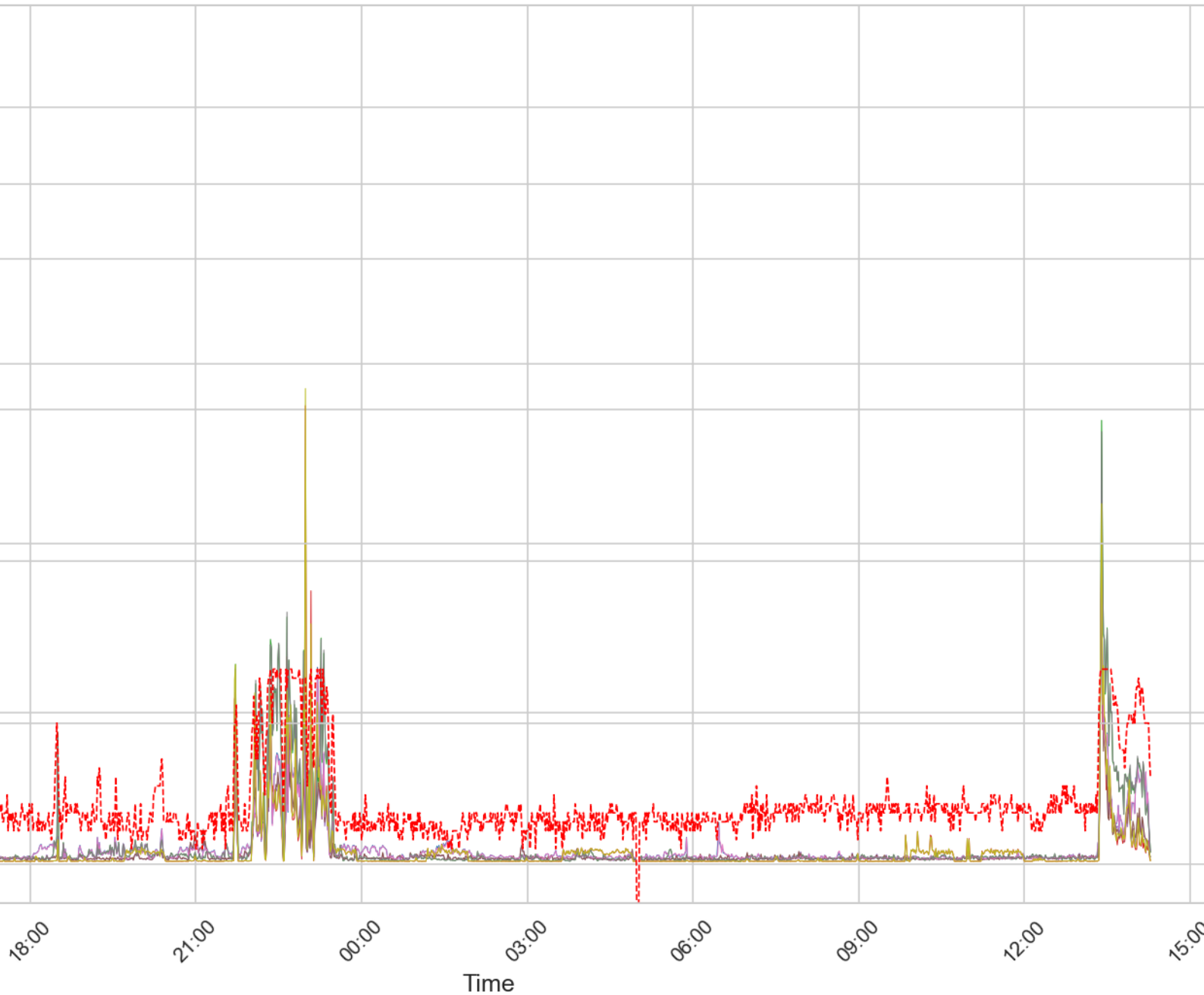
Analysis:

The chart does not show any saturation of the HIE - ISW and MPU - HIE internal paths.

6.4 Port Response Time Analysis

Port Response Time vs Write Pending Rate

Date: 2025/12/18 | Threshold: 5000 μ s



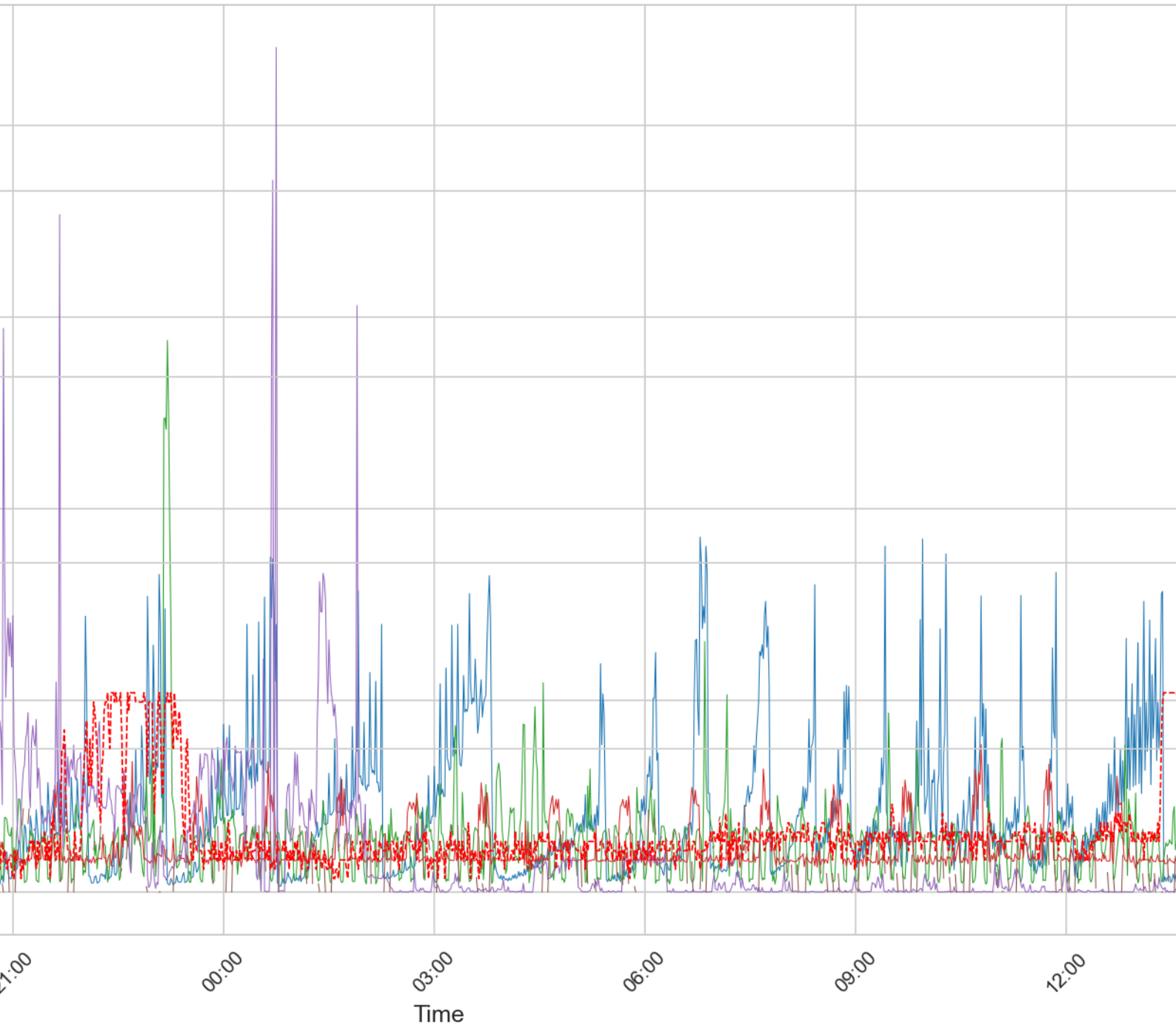
Analysis:

Port response times should be in the range of 1 ms to 3 ms. When the write pending is high the chart shows that the Port response times are above 10 ms as well.

6.5 LDEV IOPS Analysis

LDEV IOPS vs Write Pending Rate (Top 5)

Date: 2025/12/18



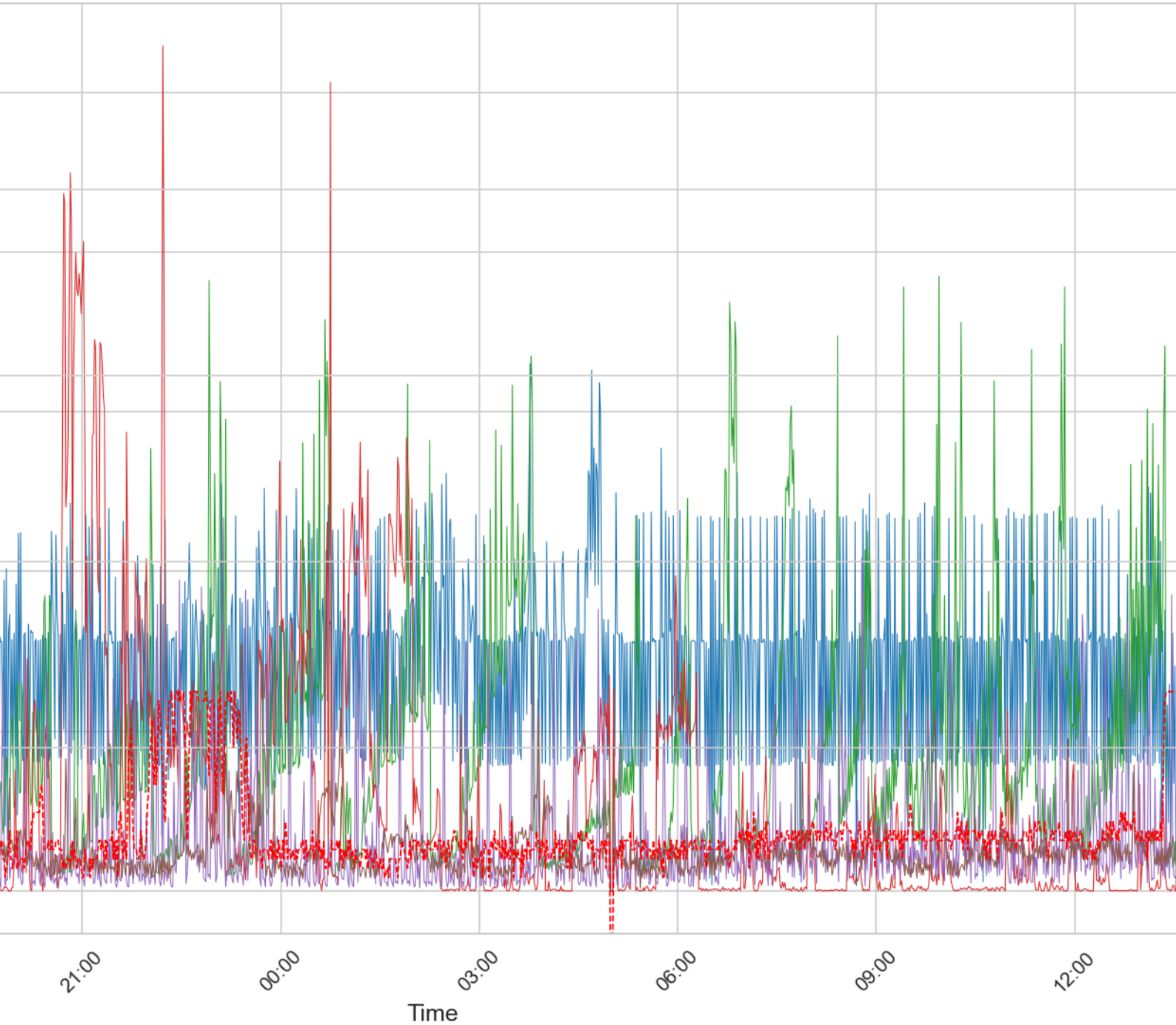
Analysis:

The chart identifies the LDEVs with high IO activity. These LDEVs probably contribute to the high write pending activity seen between 21:00 and 00:00.

6.6 LDEV Transfer Rate Analysis

LDEV Transfer Rate vs Write Pending Rate (Top 5)

Date: 2025/12/18 | Threshold: 100 MB/s



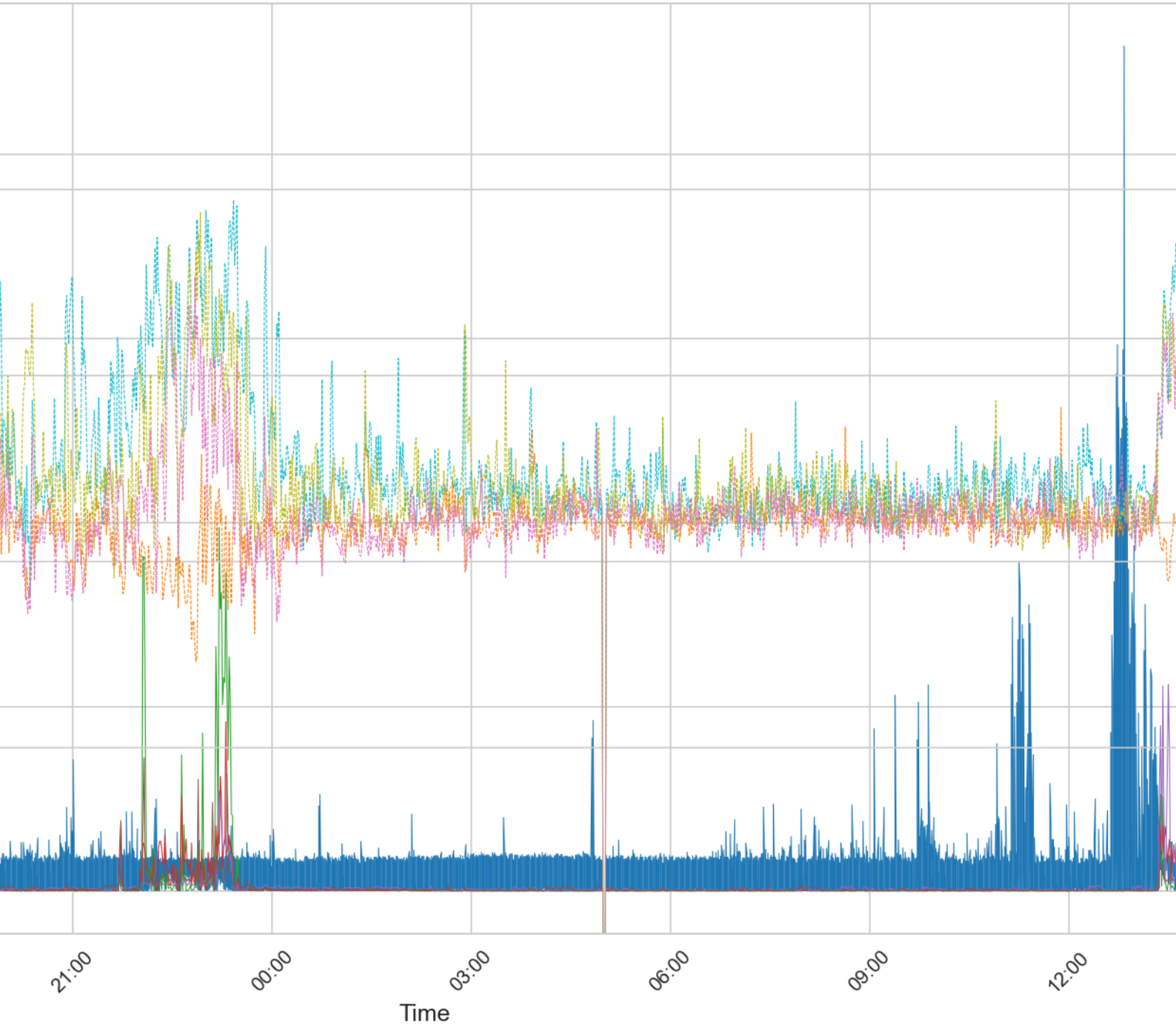
Analysis:

The LDEVs that have high transfer rate are identified in this chart. They will contribute to a noisy neighbour scenario where other LDEVs are IO starved.

6.7 LDEV Response Time Analysis

LDEV Response Time vs MPU Usage (Top 5)

Date: 2025/12/18



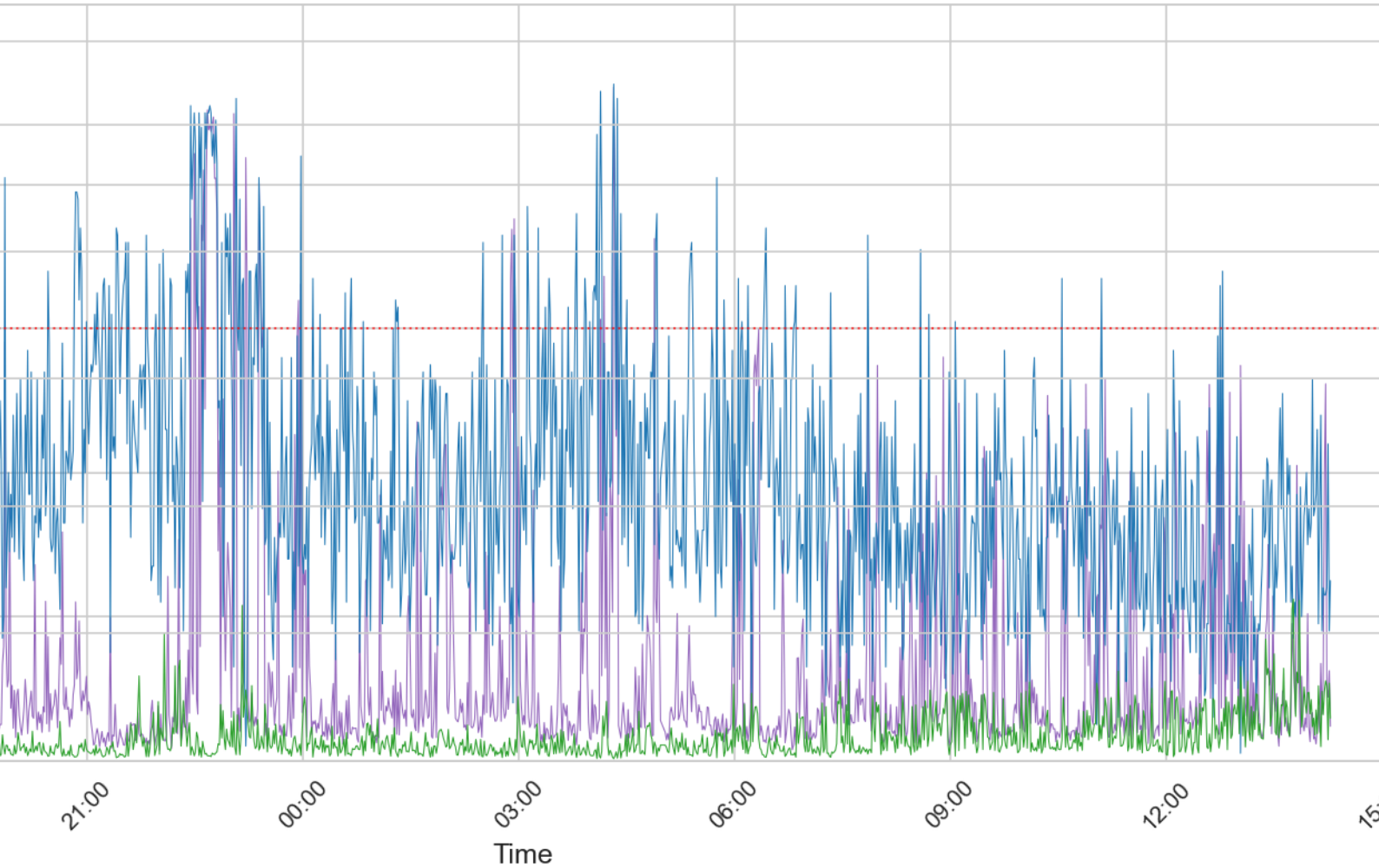
Analysis:

LDEVs that are affected by the high MPU usage and write pending between 21:00 and 00:00 are identified.

6.8 LDEV Read Hit Analysis

LDEV 00:00:D0 - Read Analysis

Date: 2025/12/18 | Avg Read%: 12.5% | Avg Read Hit: 39.0%



Analysis:

These set of charts shows the read hit%, read percentage and read response for the IO starved LDEVs identified earlier.